

PHASE 5

Project Development Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

Project Focus
AI-assisted learning resources, secure backend services, and personalized study support.

Phase Objective

To implement the planned AI StudyBuddy backend and connect its authentication, database, file handling, REST APIs, and generative AI services.

5.1 Backend Implementation

The backend is developed with Node.js and Express.js. Its modules are separated into routes, controllers, middleware, models, and utility services so individual responsibilities remain clear.

5.2 Application Entry Point

The index.js file initializes Express, loads environment variables, configures middleware, establishes the MongoDB connection, registers API routes, and starts the server after the database connection succeeds.

5.3 Database Connectivity

A dedicated database utility connects the application to MongoDB through Mongoose. The connection is established during server startup so API operations can rely on an available database.

5.4 Authentication and Authorization

- JWT access tokens are checked before protected operations are allowed.
- Authentication middleware identifies the requesting user.
- Role checks restrict administrative functions to authorized accounts.
- Passwords are protected with bcryptjs before persistence.

5.5 AI Service Integration

A dedicated Gemini utility sends appropriate prompts to the Google Gemini model and returns generated results. The service supports summaries, flashcards, quizzes, and personalized study plans.

5.6 File Upload Handling

The upload middleware accepts study-material files and validates incoming uploads before the content is processed by the rest of the application.

5.7 API Development

- Authentication endpoints support registration and login.
- Material endpoints support uploading and managing study content.
- AI endpoints generate summaries, flashcards, quizzes, and study plans.
- Administrative endpoints provide restricted management and monitoring operations.

5.8 Development Outcome

The development phase produces the connected backend required for authenticated learning workflows, persistent study resources, AI-powered generation, and administrator-level management.